04- THERMAL AND ELECTRICAL CONDUCTIVITY OF METALS

PRINCIPLE

The thermal conductivity of copper and aluminium is determined in a constant temperature gradient from the calorimetrically measured heat flow.

The electrical conductivity of copper and aluminium is determined and the Wiedmann-Franz law is tested.

TASKS

1. Determine the heat capacity of the calorimeter in a mixture experiment as a preliminary test. Measure the calefaction of water at a temperature of 0 °C in a calorimeter due to the action of the ambient temperature as a function of time.
2. To begin with, establish a constant temperature gradient in a metal rod with the use of two heat reservoirs (boiling water and ice water). After removing the pieces of ice, measure the calefaction of the cold water as a function of time.
3. Determine the electrical conductivity of copper and aluminium by recording a current-voltage characteristic line.
4. Test of the Wiedmann-Franz law.

THEORY

1. **Thermal Conductivity of Metals**

In insulating and non-metallic solids, heat is conducted entirely by phonons, meaning heat transfer can be explained through lattice vibrations. In metals, however, both electrons and phonons contribute to heat conduction. In many metals, the contribution of electrons is 10^2 orders of magnitude greater than the contribution of phonons and is therefore negligible. Therefore, in the mid-temperature region, thermal conductivity in metals is caused by the heat carried by the degrees of freedom of electrons, which can be considered as an electron gas. The Drude model and the free electron model successfully explain thermal conductivity in metals and the Wiedeman-Franz law in this context.

Below is a brief overview of the theoretical information required for the experiment. Thermal flux density is given by Fourier's law:

$$\vec{J} = -\lambda \overrightarrow{\nabla T}$$

If a temperature difference exists between different locations of a body, heat conduction occurs. In this experiment there is a one-dimensional temperature gradient along a rod. The quantity of heat dQ transported with time dt is a function of the cross-sectional area a and the temperature gradient δT/ δx perpendicular to the surface.

$$\frac{dQ}{dt} = -\lambda A . \frac{\partial T}{\partial x} \tag{1}$$

λ is the heat conductivity of the substance.

The temperature distribution in a body is generally a function of location and time and is in accordance with the Boltzmann transport equation

$$\frac{\partial T}{\partial t} = \frac{\lambda}{p.c} . \frac{\partial^2}{\partial^2} \tag{2}$$

After a time, a steady state

$$\frac{\partial T}{\partial x} = 0 \tag{3}$$

is achieved if the two ends of the metal rod having a length l are maintained at constant temperatures $T_1$and T_2, respectively, by two heat reservoirs.

Substituting equation (3) in equation (2), the following equation is obtained:

$$T(x) = \frac{T_2 - T_1}{l} . \mathrm{x} + T_1 \tag{4}$$

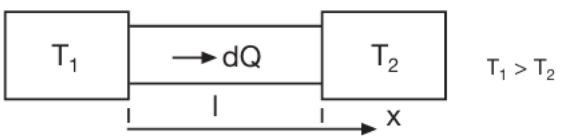
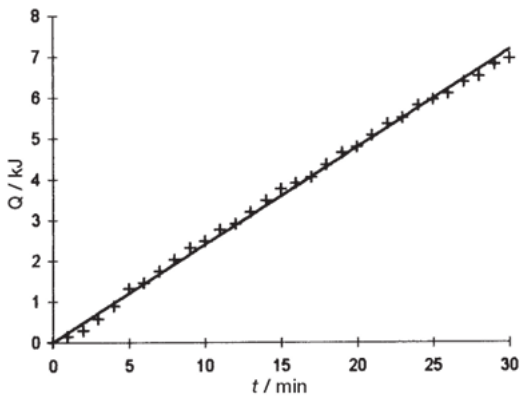


Fig. 3. Diagram: Heat of surroundings over time.



1. The heat capacity of the calorimeter is obtained from results of the mixing experiment and the following formula:

$$C_k = c_w . m_w \frac{\vartheta_w - \vartheta_M}{\vartheta_w - \vartheta_R}$$

c_w= Specific heat capacity of water

m_w = mass of the water

ϑ_w = temperature of the hot water

ϑ_m = temperature of the mixture

ϑ_R is the temperature of the room

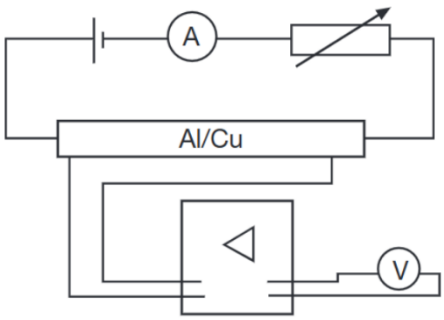
The measurement supplies a value of approximately 78 J/K ± 25%. The large variations in the results are a result of the manner in which the experiment is performed and of the experimental set-up. The addition of heat from the surroundings is calculated from the temperature increase (T of the cold water in the calorimeter.

$$\Delta Q = (c_w . m_w + C). \Delta T \tag{6}$$

where $\Delta T = T - T_0$

T_0= Temperature at time t=0

2. The heat energy supplied to the lower calorimeter can be calculated using Equation (6). The values and the change in the temperature difference on the metal rod are plotted as a function of time.



In the diagram illustrating the temperature difference, one can see that the temperature essentially remains constant. Consequently, equation 3 can be considered as having been satisfied. In order to calculate the heat energy transported by the metal rod according to Equation 1, the ambient heat fraction must be subtracted.

$$\frac{dQ_{stab}}{dt} = \frac{dQ_{gas}}{dt} = -\frac{dQ_{Umg}}{dt} \tag{7}$$

dQ/ dt for the ambient heat can be calculated from the slope of the graph in Fig. 3. dQ/dt for the entire set-up can be calculated from the slope of the graph of Q over t in Figs. 4 and 5. With the values for the length of the rod (Δx= 31.5 cm), the area (A: 4.91 · 10-4m^2) and the

averaged temperature on the metal rod, the heat conduction number can be calculated using Equation (1). The following result as the average values:

$$\lambda_{Al} = 254 \frac{W}{Km}$$

$$\lambda_{Cu} = 447 \frac{W}{Km}$$

The literature values are:

$$\lambda_{Al} = 220 \frac{W}{Km}$$

$$\lambda_{Cu} = 384 \frac{W}{Km}$$

Fig. 4a: ΔT as function of time for aluminium.

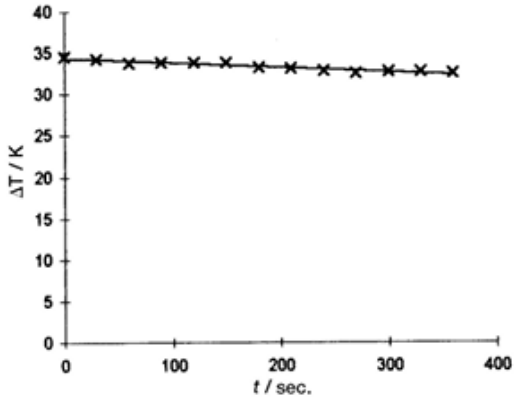
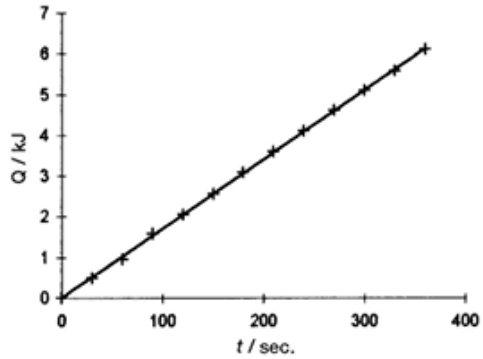


Fig. 4b: Q as a function of time for aluminium.



3. At room temperature the conduction electrons in metal have a much greater mean free path than the phonons. For this reason heat conduction in metal is primarily due to the electrons. The resulting correlation between the thermal conductivity and the electrical conductivity is established by the Wiedmann-Franz law:

$$\frac{\lambda}{\sigma} = LT \tag{8}$$

The Lorenz number L, which can be experimentally determined using Equation (8), is established by the theory of electron vapour (for temperatures above the Debye temperature) to be:

$$L = \frac{\pi^2}{3}\frac{k^2}{e^2} = 2.4.10^{-8}\ \frac{W\Omega}{K^2} \tag{9}$$

k- Universal gas constant= 1.38. $\frac{10^{-23}\ J}{K}$

E- El3mentary unit charge= 1.602. $10^{-19}\ AS$

The electrical conductivity is determined by the resistance R of the rod and its geometric dimensions (l = 0.315 m, A = 4.91 $10^{-4}m^2$).

$$\sigma = \frac{l}{A.R} \tag{10}$$

$$V = \frac{l}{\sigma A} \tag{11}$$

From Equation (8) The following values result for T= 300 K and the from the second part of the experiment.

	$R/10^{-6}\ \Omega$	$\sigma/10^{7}\ (\Omega m)^{-1}$	$L/10^{-8}\ W\Omega K^{-2}$
AL	19.6	3.27	2.5
CU	12.04	5.33	2.35

The Debye temperatures of copper and aluminium are 335 K and 419 K, respectively. Below the Debye temperature the ratio of the conductivity is smaller that given by Equation (8).

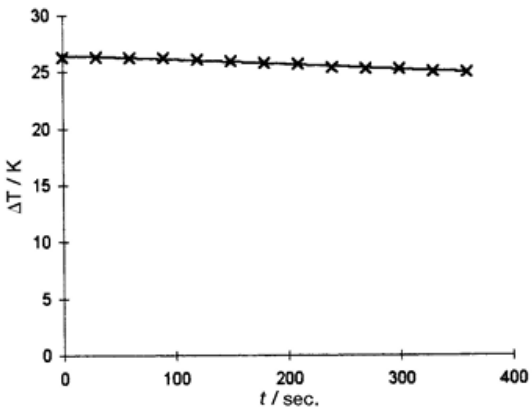


Fig. 5a: ΔT as function of time for copper.

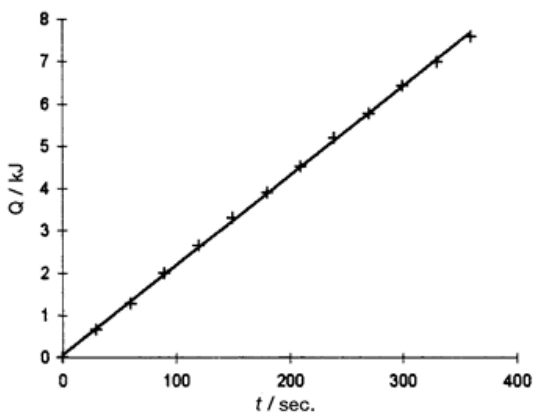


Fig. 5b: Q as function of time for copper.

Equipment for thermal measurement:

1. Calorimeter container (hot heat reservoir) 500ml
2. Calorimeter container (cold reservoir) 500ml
3. Copper/Aluminum rod (L = 0.315m, S = 4.91× 10−4m²)
4. Digital temperature gaug
5. Temperature probe (immersion type) Pt100
6. Surface temperature probes (two) Pt100

7. Heater
8. Support steel rods
9. Magnetic stirrer bar (mini)
10. Heat conductive paste
11. Connecting clamps

Equipment for electrical measurements:

1. Power Supply
2. Digital multimeters (ammeter and voltmeter)
3. Variable resistor (rheostat)
4. Amplifier
5. Connection cables
6. Copper/Aluminum rod

Set-up and Prodecure:

Part I:

Measurement of the heat capacity of the lower calorimeter

1. Weigh the calorimeter at room temperature (m).
2. Put some cold water (mass m_1) into the calorimeter container, close the container and measure its temperature (T=T_1). (To determine the mass of the water, measure the mass of the container filled with water and subtract the mass of the empty container).
3. Add some hot water (mass m_2) with a known temperature (T=T_2) to the calorimeter container.
4. Measure the temperature (T=T) when the mixture reaches thermal equilibrium.
5. Calculate the heat capacity of the calorimeter container using the following formula.

$$C_k = c_w.\frac{[m_2(T_2 - T) + m_1(T_1 - T)]}{(T - T_2)}$$

6. In this experiment, the heat capacity C=78±20 J/˚K given to the manufacturer for the calorimeter vessel can be used.

Part II:

Determining the Effect of the Environment

To find the thermal conductivity coefficient, it is necessary to find the heat flowing through the rod to the cold reservoir per unit time in steady state.

However, heat also flows from the environment to the cold reservoir during the experiment. To calculate the heat flowing from the environment to the cold reservoir, Table I is created

by following the procedure below.

1. Measure the mass of the calorimeter container.
2. Put some water (200-300 ml) into the calorimeter container.
3. Place the magnetic bar in the calorimeter container.
4. Put ice in a small cloth bag and immerse it in the water in the calorimeter container.
5. Place the calorimeter container on the magnetic stirrer and start it; this will cause the magnetic bar to rotate and mix the water, cooling it homogeneously.
6. Measure the temperature of the water-ice mixture with an immersion type temperature probe.
7. Wait until the temperature of the water-ice mixture is approximately 0°C.
8. When the temperature is approximately 0°C, remove the ice from the water in the calorimeter container and record the temperature at that moment as T0. From this point, measure the water temperature at one-minute intervals for 20 minutes and record it in Table 1.
9. After the measurements are complete, remove the magnetic bar from the water and measure the mass of the calorimeter container filled with water.
10. Subtract the mass of the container you measured earlier from this mass and record the mass of the water (m_{w1}).
11. $\Delta Q = (c_w . m_w + C).\Delta T \ , \quad \Delta T = T - T_0 \quad , i = 1, \dots .20$
12. $\Delta Q_{env.} - t$
13. $\Delta T - t$
14. Slope of $\Delta Q_{env.} - t$ gives the heat flowing from the environment to the calorimeter container per unit time $\frac{d\Delta Q_{env.}}{dt}$.

Table I:

$m_{w1} =$ gr $T_0 =$ ^{0}K			
t (mins)	T (^{0}K)	ΔT(rod)(^{0}K)	ΔQ_{total}(J)
1			
2			
3			
4			
5			
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Part III:

Determination of the thermal conductivity

1. Perform the experimental set-up according to Fig.

2. Weigh the empty, lower calorimeter.

3. Insert the insulated end of the metal rod into the upper calorimeter vessel. To improve the heat transfer, cover the end of the metal rod with heat-conduction paste.

4. Attach the metal rod to the support stand in such a manner that the lower calorimeter can be withdrawn from beneath it.

5. The height of the lower calorimeter can be changed with the aid of the supporting block. When doing so, care must be taken to ensure that the non-insulated end of the rod remains completely immersed in the cold water during the experiment.

6. The surface temperature probe must be positioned as close to the rod as possible.

7. The outermost indentations on the rod (separation: 31.5 cm) are used to measure the temperature difference in the rod. To improve the heat transfer between the rod and the surface probe, use heat-conduction paste.

8. Using an immersion heater, bring the water in the upper calorimeter to a boil, and keep it at this temperature.

9. Ensure that the upper calorimeter is well filled to avoid a drop in temperature due to contingent refilling with water.

10. Keep the water in the lower calorimeter at 0°C with the help of ice (in a gauze pouch).

11. Observe the ΔT temperature difference between two separate points on the probe (the point closest to the upper calorimeter container and the point closest to the lower calorimeter container) using surface temperature probes and a digital thermometer.

12. Measurements can begin when a constant temperature gradient (i.e., the ΔT temperature difference between the two points remains constant) is achieved (this may take 15-20 minutes to achieve).

13. When ΔT is constant, remove the ice from the cold water, record the temperature you read from the digital thermometer with the submerged probe in the lower calorimeter as T_0.

14. Simultaneously record the ΔT temperature difference from the digital thermometer.

15. For 20 minutes, at one-minute intervals, read the water temperature in the lower calorimeter container and the ΔT temperature difference on the probe from the digital thermometer and record them in Table II.

16. The temperature difference data (ΔT) obtained on the rod will be used to determine the value of $\frac{dT}{dx} = \frac{T_2 - T_1}{x_2 - x_1}$

17. When the measurements are finished, measure the mass of the lower calorimeter container together with the water inside and subtract the mass of the empty container to find the mass of the water (m_{w2}) and write it in Table 2.

18. Draw the graph of ΔT (temperature differences measured on the rod)$-t$ and fit it to a line. From this, the steady state can be seen following eq.

$$\frac{dQ_{\Delta x}}{dt} = \rho S \Delta x c_m \frac{\partial T(x,t)}{\partial t}$$

19. Calculate the heat flowing from the surroundings and through the rod to the lower calorimeter vessel using Eq. $\Delta Q = (c_w . m_w + C).\Delta T$, $\Delta T = T - T_0$ $, i = 1, \ldots .20$ for each temperature value (ΔQtpm) and write it in Table II.

20. Draw the graph of $\Delta Q_{tot.} - t$. The slope of this graph gives the heat flowing from the surroundings and through the rod to the lower calorimeter vessel per unit time.

21. Calculate the thermal conductivity coefficient ε (for copper/aluminum) using the following equation and compare it with the value in the literature:

22. $$\frac{dQ}{dt} = \frac{dQ_{tot.}}{dt} - \frac{dQ_{env.}}{dt} = -\lambda A . \frac{\partial T}{\partial x}$$

23. For dT, calculate the arithmetic mean of the temperature differences ΔT measured across the rod, and for $dx = x_2 - x_1$, calculate the distance between the two surface probes on the rod.

Table II:

m_{w2} = gr		T_0 = ^{0}K		
t(min)	T (^{0}K)	ΔT(rod)(^{0}K)	ΔQ_{tot}(J)	$\Delta T = T - T_0$(^{0}K)
1				
2				

3				
4				
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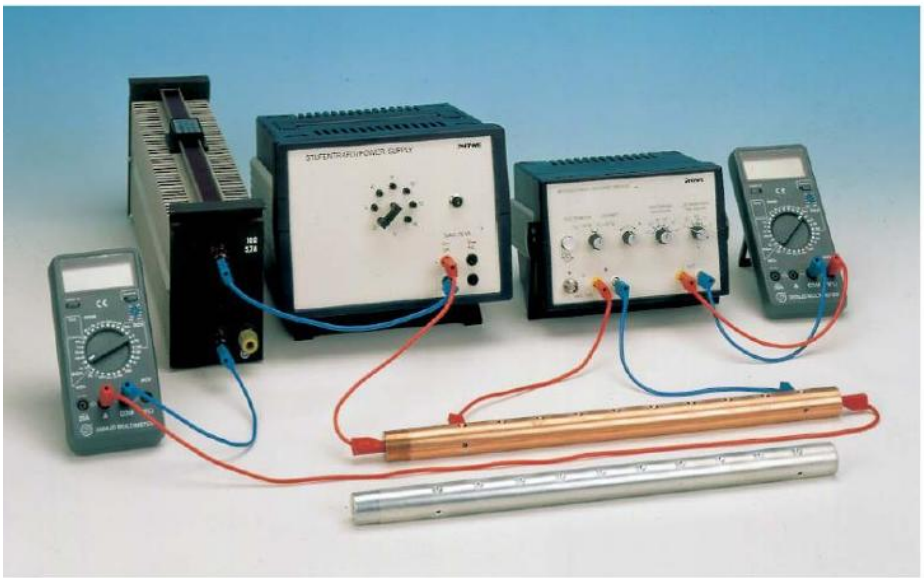
Part IV:

Measurement of the electrical conductivity.

1. Perform the experimental set-up according to the circuit diagram in Fig. 2 (set-up in accordance with a 4-conductor measuring method).



2. Set the voltage on the variable transformer to 6 V.
3. The amplifier must calibrated to 0 in a voltage-free state to avoid a collapse of the output voltage.
4. Select the amplifier settings as follows:
   Input: Low Drift
   Amplification: 10^4
   Time constant: 0
5. Set the rheostat to its maximum value and slowly decrease the value during the experiment.
6. Read and note the values for current and voltage
7. The resistance, and thus the electrical conductivity, can be determined from the measured values.
8. Determine the resistance and electrical conductivity values from the experimental data using Eqs. (10) and (11) respectively and write them in Table III. The length of the Al/Cu rod is L=0.315m and the cross-sectional area is $A=4.01\times10^{-4}m^2$.

Part V:

Testing the Wiedemann-Franz Law

Write the thermal and electrical conductivity values you found in Eq. (10) and determine the Lorentz constant L for Al/Cu metal at room temperature (300˚K) and compare it with the value given in Table IV.

Table III:

V (Volt)	I (Amp.)	R (Ω)	σ ($\Omega^{-1}m^{-1}$)

Table IV:

	$R[10^{-6}\Omega]$	$\sigma[10^{7}(\Omega m)^{-1}]$	$L[10^{-8}W\Omega K^{-2}]$
Al	19.60	3.27	2.50
Cu	12.04	5.33	2.35

REFERENCES

1. PHYWE series of publications, Laboratory Experiments